Pseudocode:

Surface blast calculator

1. input(“blasting diameter”)
2. input(“in-hole density”)
3. input("spacing to burden ratio”)
4. input(“bench height”)
5. input(“type of rock”)
6. if rock type is “hard”
       then pf=0.7
   elif rock type is “medium”
      then pf =0.45
   elif rock type is “soft”
      then pf=0.3
   elif rock type is “very soft”
      then pf=0.15
7. stemming length
   SL= 20*BD
   Print(SL)
8. Burden
   M= (p*(BD^2))/1273
   B=sqrt((M*(BH-CL))/a*PF*BH)
   Print(B)
9. Actual Powder factor
   Actual_PF= (BH-SL+U)*M/(B*S*BH)

Fragmentation calculation

1. input(“bench height”)
2. input(“blasting diameter”)
3. input("stemming length”)
4. input("in-hole density”)
5.input("Relative efficient energy of explosive)
6. input("rock factor”)
7. CL = BH - SL
8. Q = ((ρ * D**2)/1273)*CL
9. X50 = A*(K)*-0.8*Q0.167(115/E)**0.633

10. print(X50)